# Reshma S

Email: reshmas.regal@gmail.com | Phone: +91-8610949201 | LinkedIn: linkedin.com/in/reshma-subramanian

## PROFESSIONAL SUMMARY

• 8+ years of industry experience in ASIC physical design implementation spanning advanced Intel nodes and commercial foundry processes, with a consistent record of on-time tape-out delivery.
• Joined Samsung Semiconductor (May 2025) working on the cHBM Nemosyne project (Samsung N4), owning physical implementation from floorplanning through PV and EMIR sign-off.
• Worked on 7 successful tape-outs (Intel 7nm, 3nm, and 18A) delivering design databases ahead of deadlines at different exit milestones.
• Worked on multiple block-level design convergence across TSMC 130nm, 40nm, 28nm, GF55, UMC, and advanced Intel technology nodes.
• Experience in converging block-level designs: Synthesis, Floorplanning (FP), Place & Route (PNR), Clock Tree Synthesis (CTS), Static Timing Analysis (STA) & Engineering Change Orders (ECO).
• Experience using industry-standard EDA tools with an automation mindset to reduce TAT.
• Recognized for innovative problem-solving and a proactive approach to technical challenges.

## TECHNICAL SKILLS

**Tools:** FC, ICC2, ICC, Innovus, Calibre, RedHawk, RedHawk-SC
**Languages:** Unix, Tcl
**Foundries:** Samsung (N4), Intel Foundry, TSMC, UMC, GF
**Core Competencies:** Synthesis, Floorplanning, PnR, CTS, ECO, STA, PV, RedHawk Analysis

## PROFESSIONAL EXPERIENCE

### Physical Design Engineer | Samsung Semiconductor India Research | May 2025 – Present

*Samsung N4 Process · Project: cHBM – Nemosyne*

• Owned end-to-end physical implementation of block MGMTSS (588 memory instances) from floorplanning through full sign-off including CTS and P&R.;
• Responsible for sub-system HBMCSS physical implementation and sign-off.
• Resolved a critical floorplanning challenge in MGMTSS: high memory density created severe timing pressure; devised placement strategy to achieve timing closure.
• Addressed congestion issues in HBMCSS caused by long-hop nets through targeted floorplan and pin-assignment iterations to reduce routing detours and hotspots.
• Completed physical verification sign-off: DRC, LVS, and ERC (Calibre).
• Performed EMIR sign-off using RedHawk-SC, ensuring electromigration and IR-drop compliance.

### Physical Design Engineer | Intel Foundry Services | Sep 2021 – Apr 2025

*Intel 7nm, 3nm, 18A*

• Owned and delivered over 8 block-level and top-level designs, including NOC, CMN_tile, MXP, GIC, SWITCH, Channel blocks, and HBI (3DIC).
• Automated placement and routing for ~40K pins, reducing floorplan creation time by 85%.
• Conducted early PDN analysis (IRAP) and TSV planning for optimal block placement.
• Delivered successful tape-outs for Intel 7nm, 3nm, and 18A projects, ensuring design quality and timing convergence.
• Rewarded for finalizing a CTS structure for the Intel3 project after multiple experimentations.
• Recommended the 210H STD cell library for GVF IO die, achieving 5–7% area reduction for <1.5GHz designs with no PPA or DRC impact.
• Presented RTL-to-Floorplan (RTL2FP) flow, providing key insights for hierarchical block/SoC design.
• Spearheaded end-to-end implementation of high-bandwidth Network-on-Chip (NoC) architecture.
• Designed and implemented Hybrid Bond Interface (HBI) for advanced 3DIC chip projects.

### Physical Design Engineer | Microchip Technologies | Dec 2017 – Aug 2021

*TSMC 40nm, UMC 65nm*

• Led power planning and physical verification for TSMC 40nm automotive Ethernet chip (Carinae); owned pad-ring placement, PG net derivation, PCF generation, and seal-ring integration.
• Conducted static and dynamic power analysis (functional and scan modes, VCD and vectorless) in RedHawk; identified and resolved a cross-project Electromigration root cause recognised by senior management.

• Designed and integrated pre-routed analog blocks (76 sensitive signals, 7+ analog blocks) on Innovus for the Wolverhampton MCU32 65nm project, meeting complex resistance constraints verified by analog engineers.
• Generated inputs for RedHawk ACE including memory simulation and IP profiling (Sim2IPROF).
• Implemented GDS post-processing: metal/OD-PO fill, seal ring, Mask-ID and Rev-ID using Calibre.
• Contributed to Trafford2 project (A0 and B0 revisions), ensuring quality and performance.

## EDUCATION

M.Tech in Microelectronics | BITS Pilani, WILP | 2020 – 2022 | CGPA: 9.0/10
B.E. in Electrical and Electronics Engineering | Thiagarajar College of Engineering | 2014 – 2018 | CGPA: 8.6/10

## KEY ACHIEVEMENTS

• Received multiple awards including Fearless Innovation Awards for quality delivery and automation.
• Improved design efficiency by recommending optimised STD cell libraries, reducing area by 7%.
• Automated pin placement for ~40K pins and routing of ~28K CHI signals, reducing floorplan time from a week to a day and improving MXP timing convergence.
• Delivered impactful scripts reducing turnaround time across multiple workflows.
• Resolved critical Electromigration issues, enhancing chip reliability across technology nodes.

## PROJECTS

• cHBM Nemosyne (Samsung N4): End-to-end block/sub-system implementation — MGMTSS (588 memories) and HBMCSS — with full sign-off on advanced N4 node.
• Developed baseline floorplans and successfully integrated MXP blocks for 3DIC projects.
• Led top-level floorplanning for COD, ensuring optimal chip design and layout.
• Implemented ECO for 18A Kalamos test chip, optimising design and functionality.
• Managed bottom-up implementation of GIC block, collaborating with Library and RTL teams.
• Led full-chip physical verification for Wolverhampton including DRC, LVS, ERC, and PERC checks.
• Worked on automotive-grade semiconductor chips, ensuring compliance with industry standards.
• Developed an end-to-end wearable solution for real-time monitoring of elderly patients (accelerometer data streamed over Wi-Fi to a web application).